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STRAWMAN - Think Twice: Does Repeating Reasoning Insights Increase Performance?

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Abstract: [OPEN: write last, once H3 and H4 are settled] Skeleton, one sentence each: (i) Distilled reasoning insights improve accuracy but are always injected in one fixed position. (ii) Prompt repetition improves non-reasoning accuracy through the attention channel. (iii) Nobody has combined the two, so the ordering of an insight and the interaction of ordering with repetition are unstudied. (iv) We evaluate seven non-reasoning models across three families and three parameter buckets on 1,000 paired math items under the conditions `q`, `qq`, `qi`, `iq`, `qiq`, `qiqi`, `iqiq`. (v) Headline: problem-first beats insight-first, and repetition recovers the insight-first deficit, which points to a single causal-masking mechanism behind both effects.

Keywords: prompt composition, reasoning insights, prompt repetition, ordering effects, non-reasoning language models

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1 Introduction

Two unconnected lines of research.

- Chain-of-thought elicits multi-step reasoning (Wei et al., 2022), but the strategy is re-derived from scratch on every item.
- Experts do not re-derive; they recall reusable skills. This motivates distilled insights: skill cards (Zhao et al., 2026), insight extraction from demonstrations (Wang et al., 2025a), thought templates (Yang et al., 2024).
- Distilled insights raise accuracy by injecting the transferable strategy while avoiding the verbatim copying that raw demonstrations induce (Wang et al., 2025a), and save tokens at inference because the model does not re-derive the strategy (Zhao et al., 2026).
- Independently, prompt repetition raises the accuracy of non-reasoning models (Leviathan et al., 2025): tokens in the second copy can attend over the full first pass, which compensates for unidirectional causal attention (Springer et al., 2025).

Gap.

- No work combines the two streams.
- Every insight paper fixes one composition, almost always question then insight, and never justifies it.
- Ordering is known to matter strongly elsewhere (Chen et al., 2024, Wu et al., 2025), so the fixed choice is an unexamined confound.
- Repetition has been proposed as a remedy for bad context ordering (Anonymous, 2024a), but never for insights and never for non-reasoning models.

Research questions.

- **RQ1:** Do distilled insights improve accuracy for non-reasoning models, which cannot self-generate the strategy?
- **RQ2:** Does the ordering of question and insight change the effect of the insight?
- **RQ3:** Does repeating question and insight beat repeating the question alone, and does repetition interact with ordering?

1.1 Contributions

- First controlled study of prompt composition over question and insight blocks for non-reasoning models, isolating ordering from repetition on identical items.
- Evidence that the ordering of a distilled insight is a first-order design decision, not a stylistic one. [OPEN: effect size once H3 table is filled]

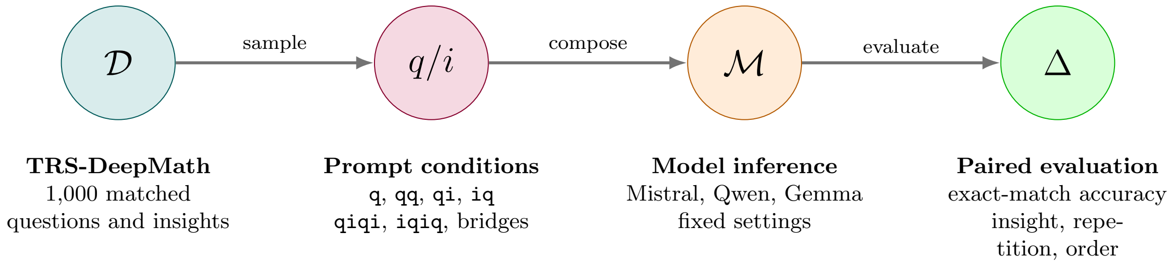

Figure 1: Experimental process. The same matched questions are evaluated under different prompt compositions. Model and decoding settings remain fixed while the order and repetition of the question and insight change.

- Evidence that repetition recovers the deficit of the weak ordering, which subsumes two apparently separate effects under one attention-level mechanism.
- A cost analysis: whether `qiqi` is worth twice the prompt.

2 Related Work

2.1 Reusable skills and distilled insights

- **TRS** (Zhao et al., 2026): offline distillation of traces into structured skill cards (Trigger / Do / Avoid / Check / Risk), online top-k retrieval, training-free and black-box. Key finding for us: structure plus a large library is required; raw examples, raw traces, and free-form summaries all underperform, so it is not the case that any relevant context helps. Source of our dataset, which removes the retrieval confound. Reasoning models only.
- **I2S** (Wang et al., 2025a): raw demonstrations cause semantic misguidance and strategy-transfer failure; converting them into explicit insights fixes this. Reasoning models only. Ordering is fixed to question then insight without discussion.
- **Adjacent work**, one line each, see Table 7: BoT (Yang et al., 2024), RoT (Anonymous, 2025c), ArcMemo (Anonymous, 2025b), Metacognitive Reuse (Didolkar et al., 2025), ReasonFlux (Yang et al., 2025), ExpeL (Zhao et al., 2023), Reflexion (Shinn et al., 2023), Self-Discover (Zhou et al., 2024), Analogical Prompting (Yasunaga et al., 2024).
- Common to all: the insight is placed after the question, once. [NOTE: this is the sentence that opens our gap]

2.2 Why insights work, and why non-reasoning models are different

- Reasoning models thrash: unnecessary switching between reasoning paths correlates with lower accuracy (Wang et al., 2025c), accuracy declines as chains grow even controlling for difficulty (Ballon et al., 2025), switches concentrate at high-entropy forking tokens (Wang et al., 2025b), and failed branches are sticky because models cannot unsee their dead ends (Feng et al., 2025).
- Insights do steer them: hints measurably shift trajectories (Anthropic, 2025), and certain sentences act as thought anchors (Anonymous, 2025a).
- But a reasoning model could self-generate that strategy. The insight mostly removes work it would have done anyway.
- Non-reasoning models hardly think about intermediate steps and rely on experience rather than strict step-by-step derivation (Anonymous, 2024b). No self-generated strategy, so the insight is genuinely new information.
- They can absorb it: in-context content aggregates into anchor tokens in shallow layers and is read out at the prediction position in deep layers (Wang et al., 2023).
- **The claim.** Reasoning: insight subtracts redundant search, tokens down, accuracy flat. Non-reasoning: insight adds a strategy, accuracy up, tokens roughly flat. This asymmetry is why our study targets non-reasoning models.
- **Costs.** Context length alone degrades performance even when the added tokens are masked (Du et al., 2025), and irrelevant context distracts (Shi et al., 2023). Repetition is therefore not free, and imperfect insights can stack noise.

2.3 Prompt repetition

- **Springer et al.** (Springer et al., 2025): repeating the input lets every token of the second occurrence attend over the full first pass, which repairs the quality deficit that causal attention imposes on early tokens. Around 5 percentage points on retrieval and similarity. Mechanistic ancestor.
- **Leviathan et al.** (Leviathan et al., 2025): same intuition, applied to downstream accuracy. Seven non-reasoning models, seven benchmarks, 47 wins

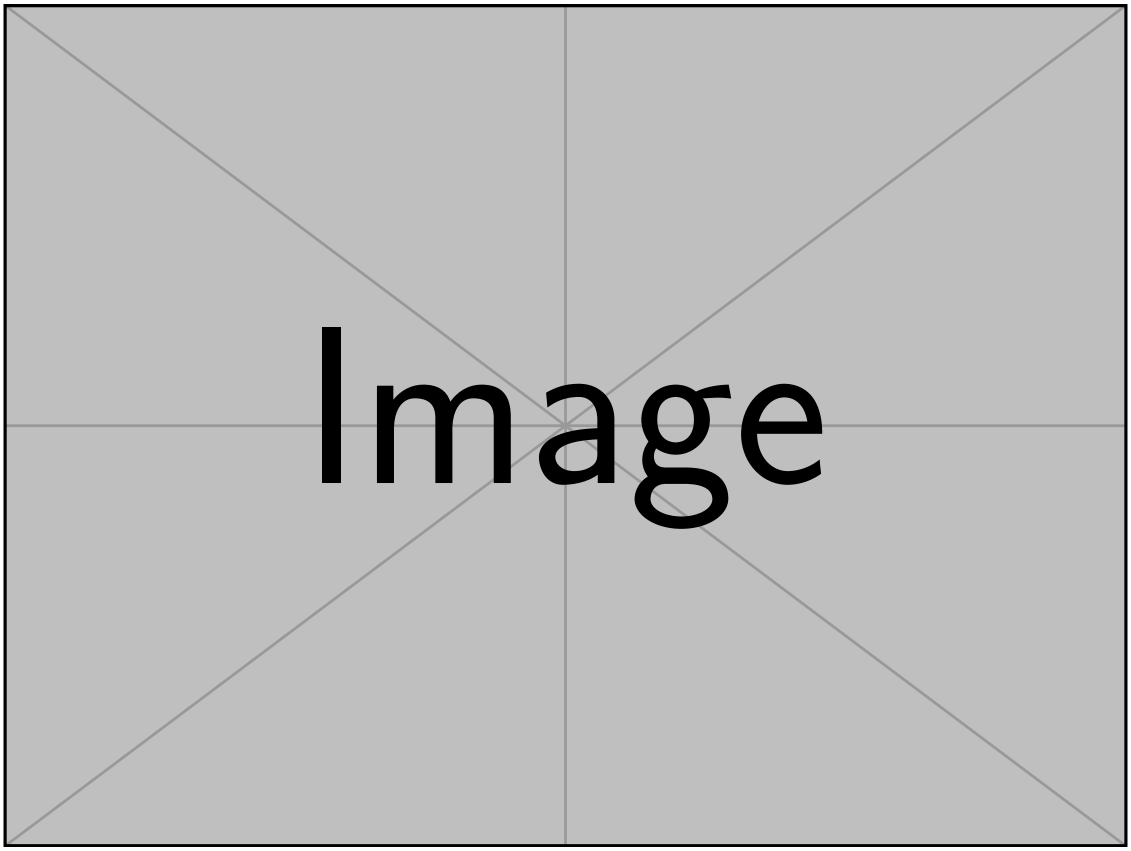

Figure 2: Prompt structure per condition. One block per condition (`q`, `qq`, `qi`, `iq`, `qiq`, `qiqi`, `iqiq`), showing that only the arrangement and multiplicity of the two blocks varies; question text, insight text, model, and decoding stay fixed. [OPEN: redraw the presentation figure in TikZ]

and 0 losses out of 70 comparisons. Ablations: repetition beats baseline and beats a length-matched padding control, so the effect is content, not length.

- Crucially, they already report an ordering interaction: gains are small with question-first and large with options-first, that is, largest when the earlier block was encoded without the later block in context. This is the pattern we predict for insight-first versus problem-first, and our closest precedent.

- **RE2** (Xu et al., 2024): re-reading the question, folded into CoT, PS, and PAL, across 14 datasets and 112 experiments, including non-IFT pretrained models. Shows repetition survives combination with other prompt structure. [NOTE: our earlier reading that RE2 contradicts Leviathan was wrong; the 5/70 figure is Leviathan's thinking-step-by-step ablation subset, not their main result. The two papers agree.]

- **CoRe** (Anonymous, 2024a): repetition proposed as a remedy for misordered context in multi-hop QA; the advantage over the no-repetition baseline grows as supporting documents move toward the middle. Direct precedent for H4, in a different domain.

- **Exploring LLM Robustness to Repeated Questions** (Shaier et al., 2025): question placed both before and after the context. Closest structural precedent to `qiq`.

2.4 Ordering effects

- **Premise Order Matters** (Chen et al., 2024): permuting premises drops accuracy by more than 30 percentage points; performance is best when premise order matches the intermediate reasoning steps; same degradation on R-GSM. Closest to our setting because it is multi-step reasoning rather than an MCQ artifact.

- **On the Emergence of Position Bias** (Wu et al., 2025): proves that causal masking biases attention toward earlier positions as depth increases. This is the formal bridge: the same mechanism that produces the ordering effect motivates repetition as its remedy.

- **Shaier et al.** (Shaier et al., 2024): context before question raises accuracy by up to 31 percentage points in reading comprehension. Cite for the existence of the effect, not as an analogue: the domain differs and the direction is opposite to ours. [NOTE: say this explicitly rather than hiding it]

- **Option-order bias** (Pezeshkpour and Hruschka, 2024, Zheng et al., 2024): reordering MCQ options swings accuracy substantially; driven by token-level priors over option identifiers. Evidence of pervasive order sensitivity, but about label bias rather than content ordering.
- **Set-based prompting** (McIlroy-Young et al., 2024): architectural remedy that modifies positional encodings and attention masks. Contrast case: infeasible for black-box models and it introduces side effects, unlike prompt-only remedies.

2.5 Positioning

- What no prior work covers: the ordering of a distilled reasoning insight for non-reasoning models, and whether prompt repetition specifically recovers the deficit of an insight-first prompt.
- Our study is the intersection: insight-ordering × repetition, on models where the insight is informative rather than redundant.

3 Hypotheses

- **H1 (insight effect, partly established).** Adding a relevant insight raises the accuracy of non-reasoning models on math. Rationale: the model cannot self-generate the strategy. Contrast: `q` versus `qi` and `iq`. See Table 1.
- **H2 (repetition effect).** Repeating the compound block raises accuracy beyond a single insight prompt, and beyond question repetition alone. Contrast: `qi` versus `qiqi`, `iq` versus `iqiq`), both benchmarked against `q` versus `qq`.) See Table 2.
- **H3 (ordering effect, dominant finding so far).** Order matters: problem-first outperforms insight-first. See Table 3.
- **H4 (repetition as an ordering remedy).** Repetition rescues the weak ordering because `iqiq` contains a `qi` substring. Prediction: $\mathrm{acc}(\texttt{qi}) \approx \mathrm{acc}(\texttt{iqi}) \approx \mathrm{acc}(\texttt{iqiq}) > \mathrm{acc}(\texttt{iq})$. If it holds, H2 and H3 are one mechanism, not two. See Table 4.
- **H5 (The second insight carries value).** The repetition gain may be entirely question repetition. Contrast: `qqi` and `qiq` against `qiqi` on one or two models. See Table 5.

Underlying mechanism (single sentence for the whole paper). Repetition operates in the delivery channel, not the information channel: the first insight already supplies the strategy, so a second copy can only matter if the model failed to attend to the first, which is exactly the non-reasoning mechanism of Leviathan et al. (2025) and predicts a larger gain for insight-first prompts, where the insight is encoded without the question in context.

4 Method

4.1 Conditions

- Baselines: `q`, `qq`.
- Insight, single: `qi`, `iq`.
- Repetition: `qiq`, `qiqi`, `iqi`, `iqiq`.
- Controls: `qqi` (question repetition with a single insight), irrelevant insight of matched length (content versus length), padding control following Leviathan et al. (2025).
- Only the arrangement varies. Question text, insight text, system prompt, model, and decoding are fixed. See Figure 2.

4.2 Metrics

- Accuracy: exact match between the extracted final answer and the gold answer, acc = correct/total.
- Order effect: $\Delta_{\mathrm{ord}} = \mathrm{acc}(\texttt{qi}) - \mathrm{acc}(\texttt{iq})$, positive means problem-first is better.
- Repetition effect: $\Delta_{\mathrm{rep}} = \mathrm{acc}(\texttt{qiqi}) - \mathrm{acc}(\texttt{qi})$, and analogously for the insight-first arm.
- Recovery: $\mathrm{acc}(\texttt{iqiq}) - \mathrm{acc}(\texttt{iq})$ compared against Δ_{ord}.
- Token usage: input, output, total, and cost per correct answer.

4.3 Statistical treatment

- Paired per item: every question is evaluated under every condition, so all contrasts are within-item.
- Transition counts per contrast: wrong to right, right to wrong, stable correct, stable wrong. [NOTE: aggregate deltas hide redistribution; the DeepSeek pilot showed equal numbers flipping both ways]
- Significance: McNemar on paired items, or paired bootstrap over items, with 95 percent confidence intervals on all deltas.
- [OPEN: decide on multiple-comparison correction across the seven models]

5 Experimental Setup

5.1 Dataset

- TRS released reasoning-skill dataset, math subset. Each item provides question q, gold answer a, and skill card i.

- Rationale: retrieval is already done, so retrieval quality is not a confound. Zhao et al. (2026) report that the injection layer backfires under poor retrieval, which is exactly what we want to hold constant.

- $n = 1{,}000$ items per model per condition, fixed seed, identical item set across all conditions and models.

5.2 Models

- Design: three families crossed with three parameter buckets (5–10B, 10–20B, 20–50B), so that inter-family and intra-family comparisons are both possible.

- Selection criterion: the family must not have been trained for reasoning at all, so that the insight is informative rather than redundant.

- Mistral Nemo 12B, Mistral Small 24B, Qwen2.5 7B, Qwen2.5 32B, Gemma 3 12B, Gemma 3 27B, Qwen3 8B with thinking disabled.

- Qwen3 8B (thinking off) is a boundary case: reasoning-trained but reasoning suppressed. Treated as a contrast, not as a clean non-reasoning model. [NOTE: it is also the one model that breaks the H3 pattern, so this matters]

- Decoding: greedy, temperature 0. [OPEN: confirm identical decoding across providers]

5.3 Robustness

- One reasoning model under the same conditions, to test the asymmetry claim directly.

- One additional seed, or prompt paraphrases if decoding is deterministic.

- One additional dataset for one family, if cheap.

- H5 decomposition (`qqi`, `qiq`) on one or two models.

6 Results

6.1 H1: Insights improve accuracy

- Table 1 shows that both insight conditions outperform the problem-only baseline `q` for every evaluated model.

- Averaged across models, accuracy increases from 43.7% under `q` to 53.6% under `qi` and 51.6% under `iq`. This corresponds to mean gains of +10.0 and +7.9 percentage points, respectively.

- The `qi` ordering performs better than `iq` for six of seven models. Mistral Nemo is the only exception, and its difference between the two orders is small.

- Overall, Table 1 provides consistent descriptive support for H1, although the magnitude of the insight effect varies across models.

6.2 H2: Repetition

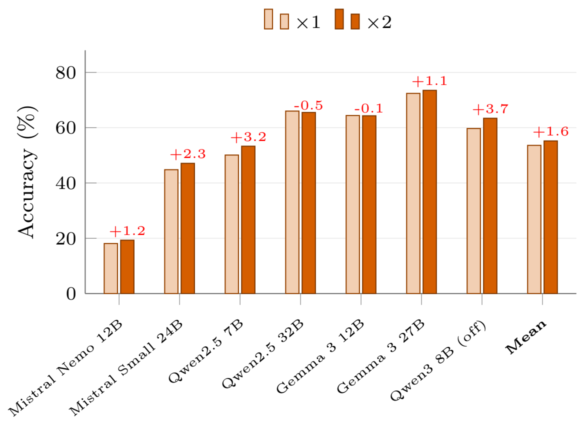

Figure 3: **Problem-first (`qi`).** Accuracy at ×1 and ×2 over the same $n = 1{,}000$ questions per condition (Table 2); labels give the repetition delta in percentage points. Repeating the full block helps in five of seven models.

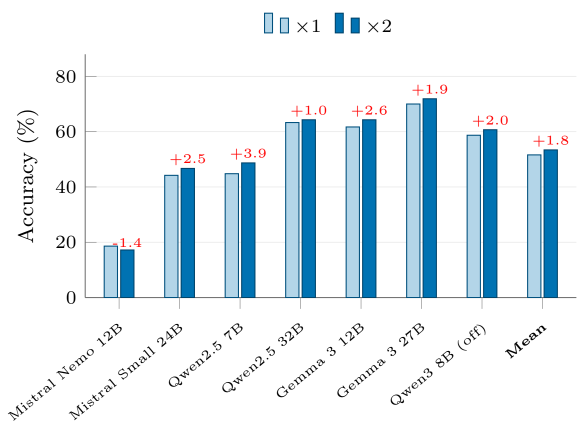

Figure 4: **Insight-first (`iq`).** Same axes and scale as Figure 3, with the insight block placed before the question. Repetition helps in six of seven models, but the ×1 level sits below problem-first for every model except Mistral Nemo 12B.

- Repetition improves accuracy in 11 of 14 model-order pairs (16 of 21 including the question-only control, Table 2). Gains are small relative to the insight effect itself: mean +1.6 pp (`qi`) and +1.8 pp (`iq`) against a ≈ 10 pp mean uplift from adding insights at all.

- The effect is not uniform. It is largest for the smaller sized models (Qwen2.5 7B: +3.2/+3.9 pp; Qwen3 8B: +3.7 pp) and vanishes or reverses for Qwen2.5 32B and Gemma 3 12B under problem-first. Mistral Nemo 12B loses accuracy under insight-first repetition (−1.4 pp), the only negative in that arm. Open: is the attenuation at

32B a capacity effect (larger models already recover the missing conditioning in one pass) or a context-length confound? Testing ×3 would separate these.

- The question-only control isolates the generic component. Repeating an arbitrary block gains +1.1 pp on its own, so the insight-specific residual is roughly +0.5 pp (`qi`) and +0.7 pp (`iq`). Most of the repetition benefit is not about the insights. The 0.2 pp residual gap between the two orders is well inside noise at n = 1,000; a significance test or a larger n is needed before this asymmetry can carry any weight.

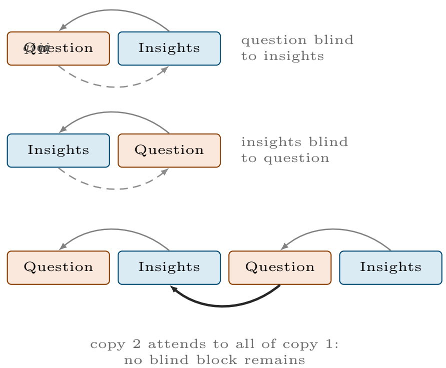

Figure 5: **Block-level causal attention.** Solid arcs show attention that causal masking permits; dashed arcs show the dependency each single-pass ordering forecloses. Under ×1 exactly one direction of conditioning is missing, and which one is missing is determined by the ordering. Repeating the full block (×2, bottom) lets every token of the second copy attend to the entire first copy, supplying the missing direction. The same construction applies to `iqiq` with the block roles exchanged.

- Causal attention permits a token to attend only to positions to its left, so content appearing later in the prompt cannot enter the contextual embedding of earlier content Springer et al. (2024), Leviathan et al. (2025). Every single-pass ordering therefore forecloses exactly one direction of conditioning (Figure 5).

- Which direction is foreclosed follows from the order: `qi` encodes the question blind to the insights, `iq` encodes the insights blind to the problem they are meant to solve. Repeating the full block lets every token of copy 2 attend to the whole of copy 1, supplying the missing direction and approximating a bidirectional encoding.

- This is the formal counterpart of the rereading intuition: a first pass fixes a representation, a second pass revises it against the full context. This account is post-hoc and consistent with the data, not evidence for the mechanism. An attention-weight or embedding-similarity probe across copy 1 and copy 2 would be needed to test it directly.

- The mechanism predicts that `iq` leaves the costlier hole open, since an insight encoded without its problem is less useful than a question encoded without its hints. Problem-first outperforms insight-first at ×1 in six of seven models (mean 53.6 vs. 51.6 pp), which is the prediction under test in the next section.

6.3 H3: Ordering

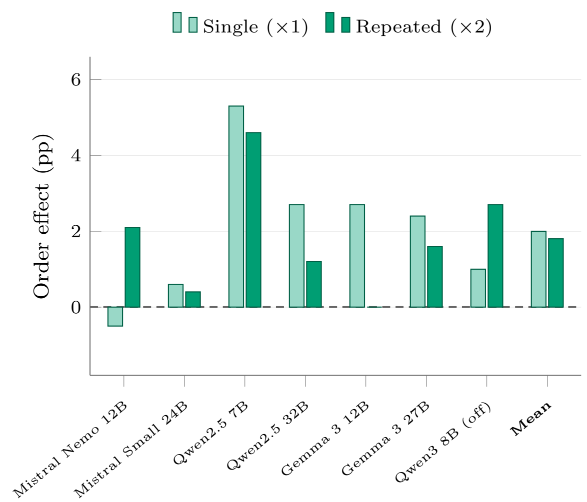

Figure 6: Order effect by model and repetition regime, over the same n = 1,000 questions per condition (Table 3). Bars show the accuracy difference between problem-first and insight-first prompts, $\mathrm{Acc}(q,i) - \mathrm{Acc}(i,q)$ for single presentation and $\mathrm{Acc}(q,i \mid q,i) - \mathrm{Acc}(i,q \mid i,q)$ for repeated presentation; positive values favor problem-first ordering. The problem-first advantage appears for nearly all models in both regimes and is largely preserved under repetition.

The ordering effect can be explained by the asymmetric conditioning induced by causal attention. Under a single presentation, the insight in qi is processed after the question and can therefore be contextualized with respect to the specific problem. In iq, by contrast, the insight is encoded before the question appears and is thus under-conditioned: although the model can access both blocks when producing the answer, the representation formed for the insight itself cannot incorporate the later question. Repetition partially repairs this asymmetry. In iqiq, the second occurrence of the insight follows the first occurrence of the question, allowing at least one insight copy to be encoded with the problem already in context. Consequently, the structural disadvantage of insight-first ordering is reduced, and the gap between qiqi and iqiq tends to be smaller than that between qi and iq. Consistent with this account, the mean order effect decreases from 2.0 percentage points under single presentation to 1.8 points under repetition, although the attenuation is modest and heterogeneous across models. This interpretation is mechanistically plausible but remains post-hoc without direct attention or representation-level analysis.

- Table 3. Positive values mean problem-first wins.
- Preliminary: positive in six of seven models at ×1, up to +5.3 pp on Qwen2.5 7B.

6.4 H4: Recovery

- Table 4 tests is whether the insight-first arm climbs to the level of `qi` once the prompt contains a `qi` substring.

The placement ablation separates the benefit of repeating the insight from the benefit of conditioning it on the question. Both tested models exhibit the same ordering, iqi > iiq > iq. Repeating the insight before the question, as in iiq, improves accuracy over iq, with a mean gain of 1.1 percentage points. Since neither insight copy can attend to the later question, this improvement is unlikely to arise from better problem-specific conditioning. Instead, repetition may increase the effective salience of the insight, making the strategy less likely to be ignored during answer generation. Placing the second insight after the question yields a larger gain: iqi improves over iq by 2.1 points on average and exceeds iiq by 1.0 point. In iqi, the final insight can be encoded with the question already in context, allowing the generic strategy to be contextualized for the specific problem. The results therefore suggest two additive components: an importance or repetition effect from presenting the insight twice, and an additional conditioning effect when the repeated insight follows the question.

6.5 H5: Does the second insight matter?

- **Test.** If adding only the second question ($q, i \mid q$ or $q \mid q, i$) matched $q, i \mid q, i$, insight repetition would be inert and the effect would reduce to known question repetition (Table **??**, Figure 7).
- **Result.** Full-pair repetition gains +3.20/+2.30/+1.20 pp (Qwen 7B / Mistral 24B / Gemma 27B), while all six question-only contrasts stay within 0.8 pp of zero ($p \geq .54$); the implied insight gain is +3.0/+2.2/+0.6 pp over the mean question-only effect.
- **Scale of the two channels.** The insight is the information channel: a single insertion is worth up to +12.8 pp, +9.9 pp on average (Table 1). The question channel is bounded by +1.1 pp for bare repetition (q to $q \mid q$, Table 2) and ≤ 0.8 pp once an insight is present. Delivery of the question is not the bottleneck: the generated solution restates the question anyway; the same boundary condition under which repetition goes inert in Leviathan et al. (2025).
- **Takeaway.** Question repetition does not reproduce the effect; the second insight copy carries it, as a repair of delivery, not as new information. It adds no content; it only raises the probability that the strategy is encoded in context of the item and read out at the answer position. Hence $q, i \mid q < q, i \mid q, i$. Only the Qwen full-pair contrast is individually significant (McNemar $p = .037$), so we read this as descriptive evidence.

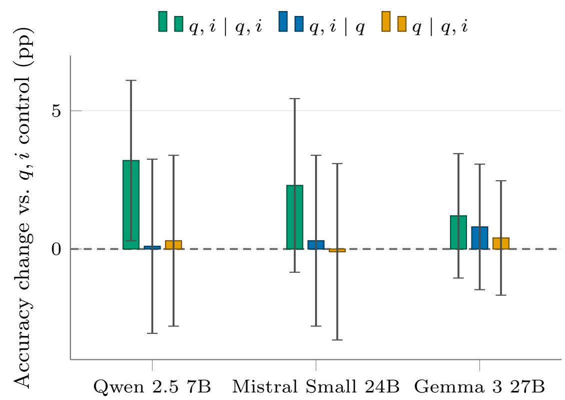

Figure 7: Full-pair repetition versus question-only repetition. Bars show the paired accuracy change relative to the same-run q, i control over $n =$ 1,000 matched questions; whiskers are paired 95% confidence intervals. Only repeating the complete question–insight pair moves accuracy; repeating the question alone, in either position, does not.

7 Discussion

- Why the order effect appears: causal masking biases attention toward earlier positions (Wu et al., 2025), so an insight placed first is encoded without the question in context and is therefore underconditioned. Mark speculative unless probed.
- Why repetition helps most where ordering hurts: the second pass restores the missing conditioning. One mechanism, two observed effects.
- Practical takeaway: if you must put the insight first, repeat; if you can choose, put the problem first and save the tokens.
- Error analysis: which items flip between conditions, and whether flips concentrate on hard items. [OPEN:]

7.1 Limitations

- Math only, one insight source (TRS cards), one retrieval regime.
- Insight quality is held constant, so we say nothing about incorrect insights, which are known to hurt (Shi et al., 2023).

- Length is a partial confound: longer context degrades performance on its own (Du et al., 2025). The padding control addresses this only for the aggregate.
- Mechanism claims are behavioral, not probed. No attention analysis.
- Open-weight non-reasoning models only, plus one suppressed-reasoning model.

8 Conclusion

[OPEN: one paragraph, written last]

References

Anonymous. Core: Repetition of misordered context. *arXiv preprint arXiv:2410.07103*, 2024a. TODO: authors and exact title.

Anonymous. Do llms really think step-by-step? *arXiv preprint arXiv:2411.15862*, 2024b. TODO: authors and exact title.

Anonymous. Thought anchors: Which llm reasoning steps matter? *arXiv preprint arXiv:2506.19143*, 2025a. TODO: authors.

Anonymous. Arcmemo: Abstract reasoning composition with lifelong llm memory. *arXiv preprint arXiv:2509.04439*, 2025b. TODO: authors.

Anonymous. Retrieval-of-thought: Efficient reasoning via reusing thoughts. *arXiv preprint arXiv:2509.21743*, 2025c. TODO: authors.

Anthropic. Reasoning models do not always say what they think. Technical report, Anthropic, 2025. TODO: verify which report.

Marthe Ballon et al. The relationship between reasoning and performance in large language models. *arXiv preprint arXiv:2502.15631*, 2025.

Xinyun Chen et al. Premise order matters in reasoning with large language models. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.08939.

Aniket Didolkar, Nicolas Ballas, Sanjeev Arora, and Anirudh Goyal. Metacognitive reuse: Turning recurring llm reasoning into concise behaviors, 2025. *URL https://arxiv. org/abs/2509.13237*, 2025.

Yufeng Du, Minyang Tian, Srikanth Ronanki, Subendhu Rongali, Sravan Bodapati, Aram Galstyan, Azton Wells, Roy Schwartz, Eliu A Huerta, and Hao Peng. Context length alone hurts llm performance despite perfect retrieval. *arXiv preprint arXiv:2510.05381*, 2025.

Feng et al. Failed branches in reasoning traces. *arXiv preprint arXiv:2509.19284*, 2025. TODO: authors and exact title.

Yaniv Leviathan et al. Prompt repetition improves language model performance. *arXiv preprint arXiv:2512.14982*, 2025. TODO: verify exact title.

Reid McIlroy-Young et al. Set-based prompting: Provably solving the language model order dependency problem. *arXiv preprint arXiv:2406.06581*, 2024. TODO: verify.

Pouya Pezeshkpour and Estevam Hruschka. Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of NAACL*, 2024. arXiv:2308.11483.

Sagi Shaier, Mario Sanz-Guerrero, and Katharina von der Wense. Asking again and again: Exploring llm robustness to repeated questions, 2025. URL `https://arxiv.org/abs/2412.07923`.

Sagi Shaier et al. Comparing template-based and template-free ordering: Context before question. In *Findings of ACL*, 2024. arXiv:2406.16779. TODO: verify title.

Freda Shi et al. Large language models can be easily distracted by irrelevant context. In *International Conference on Machine Learning (ICML)*, 2023. arXiv:2302.00093.

Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *arXiv preprint arXiv:2303.11366*, 2023.

Jacob Mitchell Springer, Suhas Kotha, Daniel Fried, Graham Neubig, and Aditi Raghunathan. Repetition improves language model embeddings. *arXiv preprint arXiv:2402.15449*, 2024. URL `https://arxiv.org/abs/2402.15449`.

Jacob Mitchell Springer et al. Repetition improves language model embeddings. *arXiv preprint arXiv:2402.15449*, 2025. Echo embeddings. TODO: verify venue.

Wang et al. From demonstrations to insights: Insight-to-solve. *arXiv preprint arXiv:2509.23196*, 2025a. TODO: verify exact title and authors.

Lean Wang et al. Label words are anchors: An information flow perspective for understanding in-context learning. In *Proceedings of EMNLP*, 2023.

Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen Yu, Gao Huang, and Junyang Lin. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for llm reasoning, 2025b. URL `https://arxiv.org/abs/2506.01939`.

Yue Wang et al. Thoughts are all over the place: On the underthinking of large reasoning models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025c. arXiv:2501.18585.

Table 1: **H1: insight effect.** Accuracy per model under the problem-only baseline `q` and the two single-insight compositions. Δ columns are percentage-point differences against `q`, computed per item on the same $n = 1{,}000$ questions. Positive means the insight helps. Significance by McNemar on paired items.

	Accuracy (%)			*Insight effect (pp)*	
Model (params)	`q`	`qi`	`iq`	`qi` − `q`	`iq` − `q`
Mistral family					
Mistral Nemo 12B	13.8	18.1	18.6	+4.3	+4.8
Mistral Small 24B	33.2	44.8	44.2	+11.6	+11.0
Qwen family					
Qwen2.5 7B	42.5	50.1	44.8	+7.6	+2.3
Qwen2.5 32B	53.2	66.0	63.3	+12.8	+10.1
Gemma family					
Gemma 3 12B	53.0	64.4	61.7	+11.4	+8.7
Gemma 3 27B	60.6	72.4	70.0	+11.8	+9.4
Reasoning-trained, thinking disabled (contrast)					
Qwen3 8B	49.5	59.7	58.7	+10.2	+9.2
Mean	43.7	53.6	51.6	+10.0	+7.9

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2201.11903.

Xinyi Wu et al. On the emergence of position bias in transformers. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.01951.

Xiaohan Xu et al. Re-reading improves reasoning in large language models. In *Proceedings of EMNLP*, 2024.

Ling Yang et al. Buffer of thoughts: Thought-augmented reasoning with large language models. *arXiv preprint arXiv:2406.04271*, 2024.

Ling Yang et al. Reasonflux: Hierarchical llm reasoning via scaling thought templates. *arXiv preprint arXiv:2502.06772*, 2025.

Michihiro Yasunaga et al. Large language models as analogical reasoners. *arXiv preprint arXiv:2310.01714*, 2024.

Zhao et al. Thinking with reasoning skills. 2026. TODO: full author list and venue.

Andrew Zhao et al. Expel: Llm agents are experiential learners. *arXiv preprint arXiv:2308.10144*, 2023.

Chujie Zheng et al. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.03882.

Pei Zhou et al. Self-discover: Large language models self-compose reasoning structures. *arXiv preprint arXiv:2402.03620*, 2024.

Table 2: **H2: repetition effect with the question-only control.** Accuracy change from ×1 to ×2 for the same $n = 1{,}000$ questions, in percentage points. The question-only arm (`q` to `qq`) isolates repetition of arbitrary content from repetition of insight-bearing content. H2 holds if the insight-bearing arms exceed the question-only control.

	Question only		*Problem-first*		*Insight-first*	
Model (params)	`q` (%)	`qq` − `q`	`qi` (%)	`qiqi` − `qi`	`iq` (%)	`iqiq` − `iq`
Mistral Nemo 12B	13.8	0.0	18.1	+1.2	18.6	-1.4
Mistral Small 24B	33.2	+0.8	44.8	+2.3	44.2	+2.5
Qwen2.5 7B	42.5	+0.5	50.1	+3.2	44.8	+3.9
Qwen2.5 32B	53.2	-1.0	66.0	-0.5	63.3	+1.0
Gemma 3 12B	53.0	+1.8	64.4	-0.1	61.7	+2.6
Gemma 3 27B	60.6	+2.0	72.4	+1.1	70.0	+1.9
Qwen3 8B (think off)	49.5	+3.4	59.7	+3.7	58.7	+2.0
Mean	43.7	+1.1	53.6	+1.6	51.6	+1.8

Table 3: **H3: order effect.** Accuracy under problem-first and insight-first ordering for the single and repeated conditions on the same $n = 1{,}000$ questions. Order effects are percentage-point differences between problem-first and insight-first prompts. Positive values indicate an advantage for problem-first ordering. Significance by McNemar on paired items.

	Single presentation			*Repeated presentation*		
Model (params)	`qi` (%)	`iq` (%)	`qi` − `iq`	`qiqi` (%)	`iqiq` (%)	`qiqi` − `iqiq`
Mistral Nemo 12B	18.1	18.6	-0.5	19.3	17.2	+2.1
Mistral Small 24B	44.8	44.2	+0.6	47.1	46.7	+0.4
Qwen2.5 7B	50.1	44.8	+5.3	53.3	48.7	+4.6
Qwen2.5 32B	66.0	63.3	+2.7	65.5	64.3	+1.2
Gemma 3 12B	64.4	61.7	+2.7	64.3	64.3	0.0
Gemma 3 27B	72.4	70.0	+2.4	73.5	71.9	+1.6
Qwen3 8B (think off)	59.7	58.7	+1.0	63.4	60.7	+2.7
Mean	53.6	51.6	+2.0	55.2	53.4	+1.8

Table 4: **H4: insight-repetition placement.** Accuracy on the same $n = 1{,}000$ questions. The baseline `iq` contains one insight. Conditions `iiq` and `iqi` add a second copy before and after the question, respectively. Differences are reported in percentage points.

	Accuracy (%)			*Insight-repetition effect (pp)*		
Model (params)	`iq`	`iiq`	`iqi`	`iiq` − `iq`	`iqi` − `iq`	`iqi` − `iiq`
Mistral Small 24B	41.2	42.2	43.1	+1.0	+1.9	+0.9
Qwen2.5 7B	46.9	48.1	49.1	+1.2	+2.2	+1.0
Mean	44.1	45.2	46.1	+1.1	+2.1	+1.0

Table 5: **H5: decomposition of the repetition gain.** All conditions contain exactly one question repetition; they differ in how many insight copies are present and where. If `qqi` matches `qiqi`, the second insight copy is inert and the gain is question repetition alone. Run on one or two models. Accuracy in percent, deltas against `qi` in percentage points.

	Accuracy (%)				
Model (params)	`qi` (ref)	`qqi`	`qiq`	`qiqi`	Insight copies that matter
Qwen2.5 7B	50.1	51.2	51.0	53.3	Second copy
Mistral 24B	44.8	44.3	44.7	47.1	Second copy
Gemma 27B	72.4	67.5	68.0	73.5	Second copy
Δ vs `qi` (pp)	0.0	-1.4	-1.2	+2.2	

Table 6: **Token efficiency.** Mean tokens per item and cost per correct answer across conditions, aggregated over models. The question is whether the accuracy gain of `qiqi` justifies roughly twice the prompt. Cost per correct answer is total tokens divided by accuracy and is the deciding metric.

Condition	Accuracy (%)	Mean input	Mean output	Mean total	Tokens per correct
`q` (baseline)	–	–	–	–	–
`qq`	–	–	–	–	–
`qi`	–	–	–	–	–
`iq`	–	–	–	–	–
`qiq`	–	–	–	–	–
`qiqi`	–	–	–	–	–
`iqiq`	–	–	–	–	–
Irrelevant insight (control)	–	–	–	–	–
Padding (control)	–	–	–	–	–

Table 7: **Related work on reusable skills and insights.** Every prior approach fixes a single composition in which the insight follows the question and appears once. Neither the ordering nor the multiplicity of the insight block has been studied, and only two entries evaluate non-reasoning backbones.

Work	Domain	Insight construction	Composition used	Backbone
TRS (Zhao et al., 2026)	Math	Structured skill cards	question + insight	Reasoning
I2S (Wang et al., 2025a)	Math, GPQA	Insight from demos	question + insight	Reasoning
BoT (Yang et al., 2024)	10 reasoning tasks	Thought templates	question + template	GPT-4
RoT (Anonymous, 2025c)	General reasoning	Thought graph	question + template	Reasoning
ArcMemo (Anonymous, 2025b)	ARC-AGI	Concept memory	question + concepts	Reasoning
Metacognitive Reuse (Didolkar et al., 2025)	Multi-step	Behavior handbook	question + behaviors	Reasoning
ReasonFlux (Yang et al., 2025)	Math	Template trajectories	question + trajectory	Reasoning
ExpeL (Zhao et al., 2023)	Agents	Experience insights	task + insights	GPT-styl
Self-Discover (Zhou et al., 2024)	BBH, MATH	Composed structure	question + structure	Non-reaso
Analogical (Yasunaga et al., 2024)	GSM8K, MATH	Self-generated exemplars	question + exemplars	Non-reaso
Ours	Math	TRS cards, reused	**ordering × repetition**	**Non-reas**

Table 8: Exact-match accuracy for all completed full-scale runs involving the eight model variants. Each populated cell contains the accuracy, in percent, for one independently executed prompt condition with $n = 1{,}000$ questions. Repeated prompt conditions in later robustness experiments are reported in separate rows and are not averaged with the main run.

		Prompt condition accuracy (%)										
Model variant	Run	$\{q\}$	$\{q \mid q\}$	$\{q, i\}$	$\{i, q\}$	$\{q, i \mid q, i\}$	$\{i, q \mid i, q\}$	$\{q, i \mid q\}$	$\{q \mid q, i\}$	$\{i, q \mid i\}$	$\{i \mid i, q\}$	Errors
Mistral Nemo 12B	Main (07-10 00:50)	13.8	13.8	18.1	18.6	19.3	17.2	–	–	–	–	0
Mistral Small 24B	Main (07-10 22:19)	33.2	34.0	44.8	44.2	47.1	46.7	–	–	–	–	0
Mistral Small 24B	Question bridge (07-14 05:18)	–	–	44.4	43.5	–	–	44.7	44.3	–	–	0
Mistral Small 24B	Insight bridge (07-15 02:35)	–	–	–	41.2	–	–	–	–	43.1	42.2	0
Qwen 2.5 7B	Main (07-10 02:02)	42.5	43.0	50.1	44.8	53.3	48.7	–	–	–	–	0
Qwen 2.5 7B	Question bridge (07-14 00:27)	–	–	50.9	46.1	–	–	51.0	51.2	–	–	0
Qwen 2.5 7B	Insight bridge (07-14 23:15)	–	–	–	47.9	–	–	–	–	50.0	49.3	0
Qwen 2.5 32B	Main (07-10 17:53)	53.2	52.3	66.0	63.3	65.4	64.3	–	–	–	–	0
Gemma 3 12B	Main (07-12 15:48)	53.0	54.8	64.4	61.7	64.3	64.3	–	–	–	–	0
Gemma 3 27B	Main (07-12 21:10)	60.6	62.6	72.3	70.0	73.5	71.9	–	–	–	–	0
Gemma 3 27B	Question bridge (07-14 23:40)	–	–	70.8	–	–	–	71.6	71.2	–	–	0
Qwen 3 8B (thinking on)	Main (07-12 16:05)	25.8	25.5	32.0	30.1	31.1	33.0	–	–	–	–	1
Qwen 3 8B (thinking off)	Main (07-13 00:21)	49.5	52.9	59.7	58.7	63.4	60.7	–	–	–	–	0

Notes. q denotes the question, i the oracle solving insight, and $\mid$ the explicit separator between prompt copies. All runs used the TRS-DeepMath oracle setting, exact-match evaluation, sample size 1,000, evaluation offset 2,000, seed 42, temperature 0.7, and a 2,048-token completion limit. "Main" denotes the original six-condition factorial run. "Question bridge" denotes the question-repetition robustness run, and "Insight bridge" denotes the insight-repetition robustness run. The two Qwen 3 rows use the same base model with the prompt suffixes `/think` and `/no_think`, respectively. "Errors" is the number of provider-error records across all populated cells in that row. Transient provider errors (timeouts, HTTP 5xx, network failures) were retried post hoc with byte-identical prompts and configuration; the column reports records that still failed after retries. Any remaining error record counts as incorrect in the reported aggregate accuracy. A dash indicates that the condition was not executed in that run.

A Additional Details

A.1 Prompt templates

- Verbatim system prompt and separator tokens per condition. [OPEN:]
- Skill card format: Trigger / Do / Avoid / Check / Risk.

A.2 Extended results

A.3 Does the model actually use the card? A blinded judging study

To separate *adoption* (does the completion apply the card's strategy?) from *execution* (does an applied strategy succeed?), we judged completions of all items that flip between q, i and $q, i \mid q, i$, the improved (wrong→right) and hurt (right→wrong) groups, for three models. Each (card, completion) pair was rated *applies / partial / does not apply* by an LLM judge under a fixed rubric, with all 1,218 judgments shuffled and blinded to condition and correctness. The hurt group is the critical control: if judged adoption merely tracks correctness, it must mirror the improved group.

Adoption rate (% yes)	Qwen 7B	Mistral 24B	Gemma 27B
improved, $\times 1 \rightarrow \times 2$	88→97	91→98	72→94
hurt (control), $\times 1 \rightarrow \times 2$	89→81	100→93	88→85
pooled, $\times 1 \rightarrow \times 2$	88.6→90.0	94.9→95.7	**79.5→90.2**
pooled sign test	$p = .72$	$p = 1.0$	$p = .016$

- **Adoption is near ceiling, even in failures.** For Qwen and Mistral, $\geq 88\%$ of *wrong* single-presentation responses already apply the card. Failures under q, i are execution failures, not attention failures; the "rescue an ignored card" channel has at most $\approx 10\%$ of items to work with.
- **For Qwen and Mistral, repetition does not change adoption.** Adoption gains on improved items are mirrored by losses on hurt items, and pooled rates are flat: judged adoption tracks the answer flip, not the condition. Their repetition gain must therefore operate through execution quality, consistent with the conditioning gradient $i, q \mid i > i \mid i, q$ (Table 4).
- **Gemma is the exception, and it proves the moderator.** Gemma adopts the card least at $\times 1$ (79.5% pooled), the only judged model with adoption headroom, and repetition genuinely raises its adoption (79.5→90.2%, $p = .016$; the hurt control does *not* mirror the improved group). The attention-rescue channel is real, but only where single-pass adoption is incomplete.
- **Adoption is not correctness.** Gemma shows the largest adoption gain and the smallest accuracy gain (+1.2 pp): a strong model that ignores the card often solves the item its own way, so rescued adoption buys little accuracy. The second copy chiefly changes *how* the model solves, not *whether* it succeeds.
- Full transition tables (wrong to right, right to wrong, stable) per contrast and model.
- Token distributions: median, p90, p95, max, near-limit rows.
- Per-bucket line charts: order effect and repetition effect against parameter count, one line per family.

A.4 Implementation details

- Serving, decoding parameters, seed, answer extraction and its failure modes.
- Compute budget and total run time. [NOTE: high n runs are the bottleneck]